Kate Salesin

Ph.D. Student, Dartmouth College

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Research Interests

I am interested in solving computer graphics problems in both the artistic and photorealistic domains, particularly capture and control of the plethora of complex materials and participating media in the world. I find problems with hands-on elements particularly exciting, such as appearance capture, fabrication, or anything that encourages going outside and exploring the world.

Education

Ph.D., Dartmouth College ..... Sept. 2018 – present
Computer Science

Advisor: **Wojciech Jarosz**

B.S. with Distinction, Cornell University ..... May 2016
Computer Science, Minor in Marine Biology

Dean's List (5/8 semesters)
Advisor: **Kavita Bala**

Publications

Combining Point and Line Samples for Direct Illumination. Katherine Salesin, Wojciech Jarosz. Computer Graphics Forum (Proceedings of EGSR), 38(4), July 2019.

Presentations

Combining Point and Line Samples for Direct Illumination. Eurographics Symposium on Rendering. Strasbourg, France. July 2019.

Awards

Runner Up, Rendering Competition ..... Nov. 2019
CS 87/187: Rendering Algorithms, Dartmouth College

Research Experience

Data Visualization Scientist ..... Summer 2018
Photonic Sentry

Photonic Sentry is a Global Good/Intellectual Ventures start-up that has created a laser that zaps mosquitoes, psyllids, and other pests out of the air. I created tools for logging, organizing, and visualizing live research data from lasers and cameras. Data typically included system statuses, bug stats, and bug flight paths.

Research Assistant ..... Summer 2014 – Spring 2015
Cornell University, Computer Science Dept.

I assisted Prof. Steve Marschner on a project to create a realistic wood texture authoring tool. I prepared wood samples and measured their reflective properties using a spherical gantry.

Teaching Experience

Instructor ..... Jan. 2020 – Mar. 2020
Dartmouth College

I designed and ran the graduate reading course for Master's in Digital Arts students, which included writing the syllabus and assignments, running discussions, and arranging presentations by guest speakers.

Teaching Assistant ..... Sept. 2018 – June 2019
Dartmouth College

CS 10: Problem Solving via Object-Oriented Programming
CS 98: Senior Design and Implementation Project

Station Leader ..... May 2019
Science Day at Dartmouth

Science Day at Dartmouth is an annual event where graduate students teach kids about their research through fun, hands-on activities. I **organized a computer graphics station** that taught kids some of the science behind their favorite movies and video games, and we acted out ray tracing in real life!

Deckhand/Educator ..... Apr. 2017 – Nov. 2017
Call of the Sea

I sailed on schooner Seaward on San Francisco Bay and along the California coast with students grade 3-12. I created interactive lesson plans and taught basic marine biology, ecology, modern and historical navigation, scientific data collection, and seamanship.

Sailing Intern ..... Nov. 2016 – Dec. 2016
Sea Education Association

I sailed on SSV Robert C. Seamans in the South Pacific near New Zealand as part of SEA Semester's Ocean Exploration program. I taught college students the fundamentals of sailing, seamanship, celestial and modern navigation, and oceanographic research on a tall ship.

Teaching Assistant ..... Aug. 2015 – May 2016
Cornell University

CS 4620: Introduction to Computer Graphics
CS 5625: Interactive Computer Graphics

Mentor Experience

Deckhand. As a deckhand on many sailing voyages, I mentored watches of 5-8 students on how to effectively care for the ship and each other while out to sea. It was especially rewarding on the longer ocean voyages to participate in forming a miniature, tight-knit community.

Teaching Assistant (CS 98). As a teaching assistant for CS 98, I mentored groups of 4-6 students on six-month software projects of their own design. I tried to inspire, motivate, and assist the students in design

choices, group dynamics, and problem solving.

Service & Professional Activities

Webmaster ..... Sept. 2018 – present
Dartmouth Visual Computing Lab Website

Judge ..... June 2019
Dartmouth Spring Hackathon

Member ..... 2015 – 2016
WICC (Women in Computing at Cornell)